

Agent365 Sampling V4 Summary Report

STATUS Validated hash, schema, and selection-stage linkage	GENERATED AT 2026-08-05T13:52:26.307194Z
POPULATION 2,800 sessions	RUNTIME 3h 11m 27s

Executive Summary

This report answers one question: under fixed exact token budgets, which sampling method best approximates census outcome rates while preserving coverage. The current run covers 2,800 sessions and completed in 3h 11m 27s. Percent tier labels are provenance labels only; exact token budgets are the authoritative comparison axis.

Experiment Design Three methods compared at matched exact budgets across 5 budget levels with persisted memberships and deterministic fills.	Outcome Winner Pattern Random won 4 of 5 budgets; MinHash won 1 of 5 budgets. Exact values are in the Outcomes tab.
IDW Verdict Embedding IDW improved 6 cells, regressed 9, tied 0.	MinHash Verdict 4,310 bucket misses were treated as novel; 0 exhaustive scans were used.

Practical Recommendation

- Random selected-only is the accuracy default for this run (4/5 budget cells won by MAE).
- MinHash won 1/5 budget cells and provides bounded collision lookup; misses are explicit novel/no-candidate events.
- Embedding led concept coverage in 3/3 medium/high budget cells in this run but adds throughput decision latency pressure.
- Post-freeze same-agent IDW is mixed (6 improved, 9 regressed, 0 tied) and should remain diagnostic or budget-specific.

Representative Exact-Budget Table

Representative budget uses 20% legacy provenance when available; otherwise median exact budget.

Reported method	Exact token budget	Selected sessions mean	Reported MAE	Fraction saved	Concept coverage
Random selected-only	263,797	566.00	0.0130	79.8%	48.3%
MinHash selected-only	263,797	508.33	0.0252	81.8%	50.5%
Full-session embedding + IDW	263,797	369.33	0.0402	86.8%	65.1%

Metric Definitions

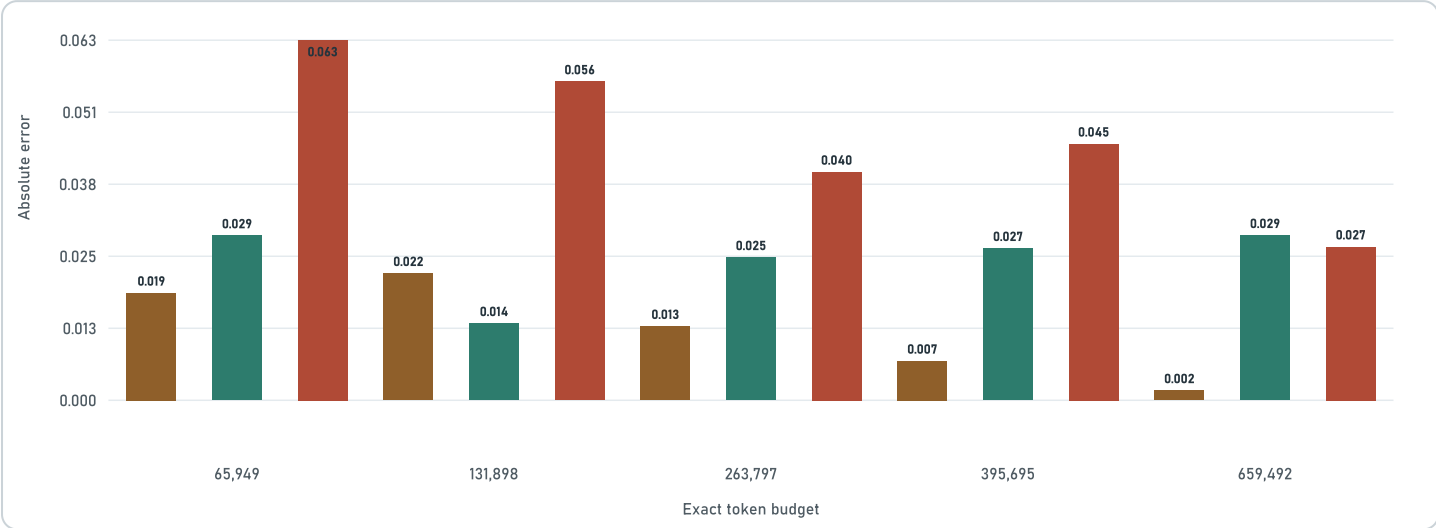
- Selected-only absolute aggregate rate error = |selected pass rate - census|.
- IDW aggregate error = |observed+imputed population mean - census|.
- Fraction saved = 1 - (selected count / population count).
- Concept coverage = covered concepts / total concepts in the run aggregate.
- Deterministic expected labels are pseudo-judge outputs; adaptive methods are not design-unbiased estimators.

Outcomes

What this shows: matched exact-budget outcome accuracy and efficiency tradeoffs. Random and MinHash report selected-only error; full-session embedding reports observed-plus-IDW population estimates.

Mean absolute error versus census by exact token budget

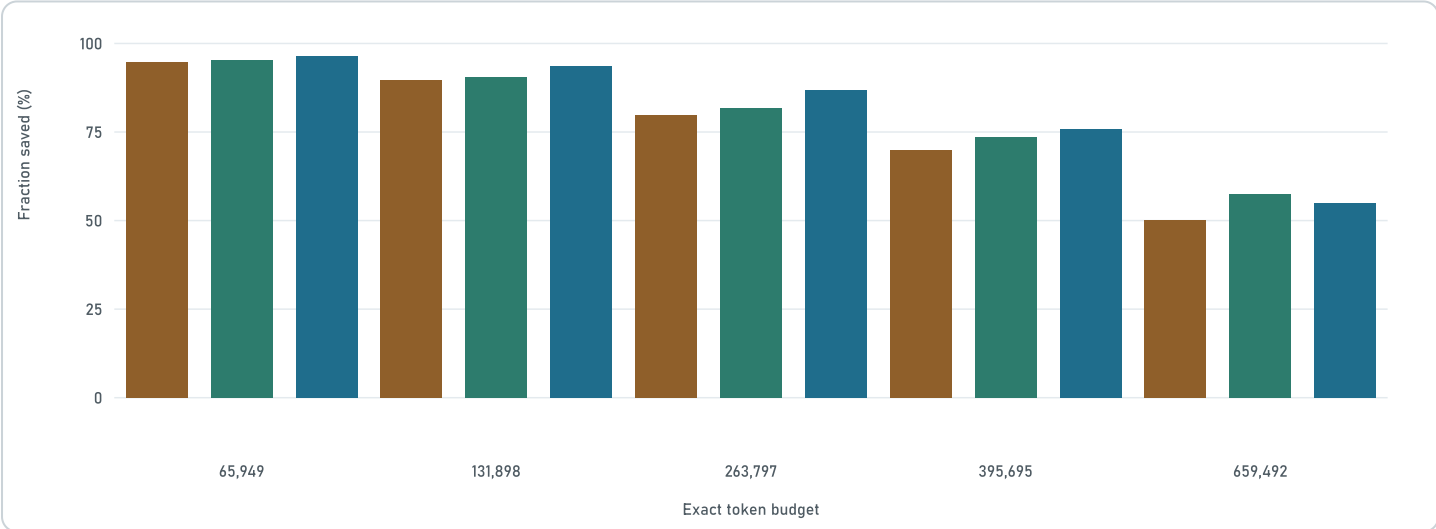
Random selected-only MinHash selected-only Full-session embedding + IDW



How to interpret it: compare methods at the same budget only. 65,949: Random selected-only lowest MAE 0.0189; 131,898: MinHash selected-only lowest MAE 0.0135; 263,797: Random selected-only lowest MAE 0.0130; 395,695: Random selected-only lowest MAE 0.0069; 659,492: Random selected-only lowest MAE 0.0017.

Fraction saved by exact token budget

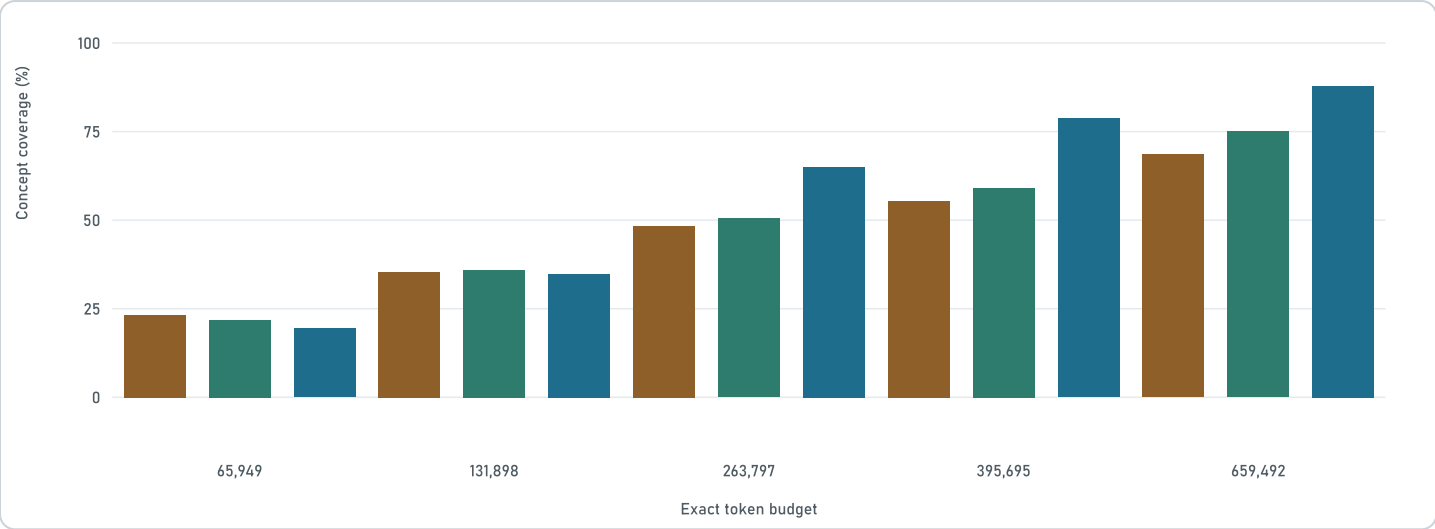
Random selected-only MinHash selected-only Full-session embedding + IDW



How to interpret it: higher means fewer sessions judged. Highest fraction saved: 65,949: Full-session embedding + IDW 96.5%; 131,898: Full-session embedding + IDW 93.7%; 263,797: Full-session embedding + IDW 86.8%; 395,695: Full-session embedding + IDW 75.8%; 659,492: MinHash selected-only 57.5%.

Concept coverage by exact token budget

Random selected-only MinHash selected-only Full-session embedding + IDW



How to interpret it: higher means broader concept surface retained by the sample. Highest concept coverage: 65,949: Random selected-only 23.2%; 131,898: MinHash selected-only 35.9%; 263,797: Full-session embedding + IDW 65.1%; 395,695: Full-session embedding + IDW 78.7%; 659,492: Full-session embedding + IDW 87.9%.

Dense Outcome Table

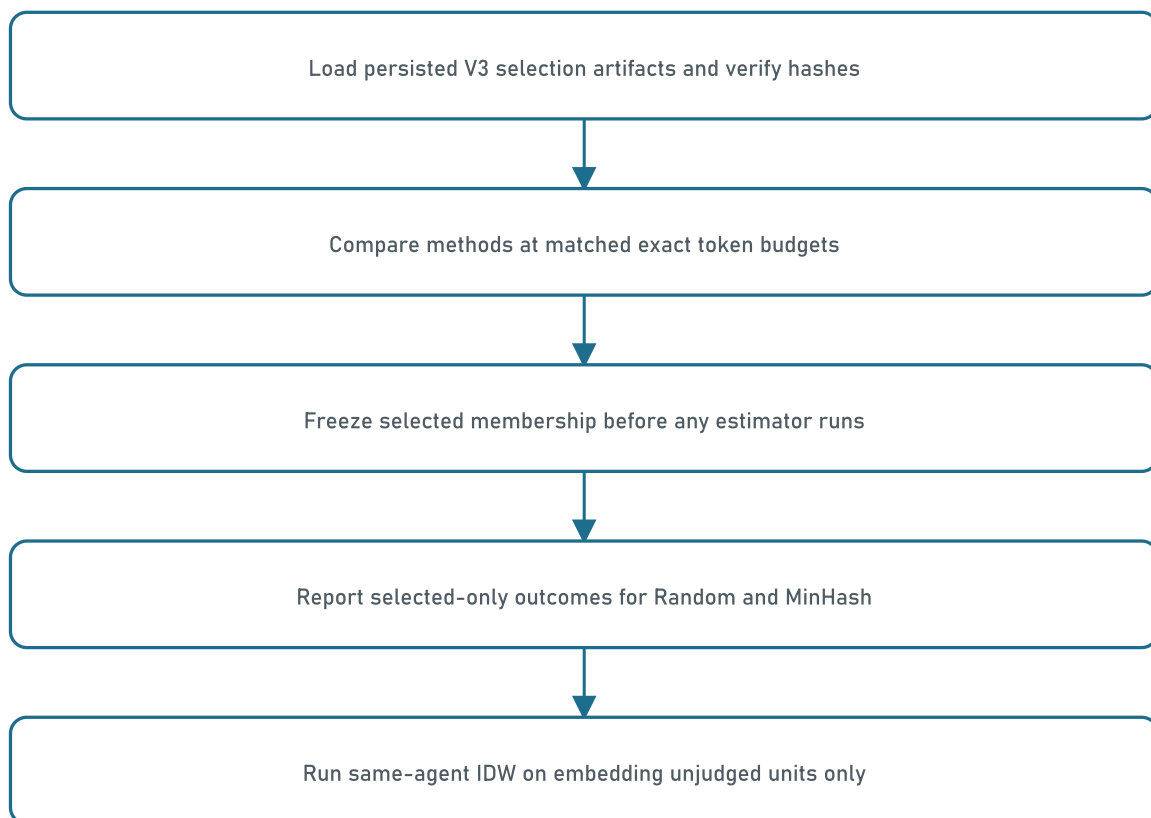
Legacy provenance tier	Exact tokens	Reported method	Selected count mean	Reported MAE	Fraction saved	Concept coverage	Note
5%	65,949	Random selected-only	147.33	0.0189	94.7%	23.2%	best reported MAE
5%	65,949	MinHash selected-only	131.67	0.0290	95.3%	21.8%	
5%	65,949	Full-session embedding + IDW	97.00	0.0634	96.5%	19.4%	
10%	131,898	Random selected-only	287.00	0.0224	89.8%	35.4%	best reported MAE
10%	131,898	MinHash selected-only	268.33	0.0135	90.4%	35.9%	
10%	131,898	Full-session embedding + IDW	176.33	0.0561	93.7%	34.8%	
20%	263,797	Random selected-only	566.00	0.0130	79.8%	48.3%	best reported MAE
20%	263,797	MinHash selected-only	508.33	0.0252	81.8%	50.5%	
20%	263,797	Full-session embedding + IDW	369.33	0.0402	86.8%	65.1%	
30%	395,695	Random selected-only	843.67	0.0069	69.9%	55.4%	best reported MAE
30%	395,695	MinHash selected-only	738.67	0.0268	73.6%	59.1%	
30%	395,695	Full-session embedding + IDW	677.00	0.0452	75.8%	78.7%	
50%	659,492	Random selected-only	1393.67	0.0017	50.2%	68.5%	best reported MAE
50%	659,492	MinHash selected-only	1191.33	0.0290	57.5%	75.0%	

Legacy provenance tier	Exact tokens	Reported method	Selected count mean	Reported MAE	Fraction saved	Concept coverage	Note
50%	659,492	Full-session embedding + IDW	1263.00	0.0270	54.9%	87.9%	

Sampling Methods

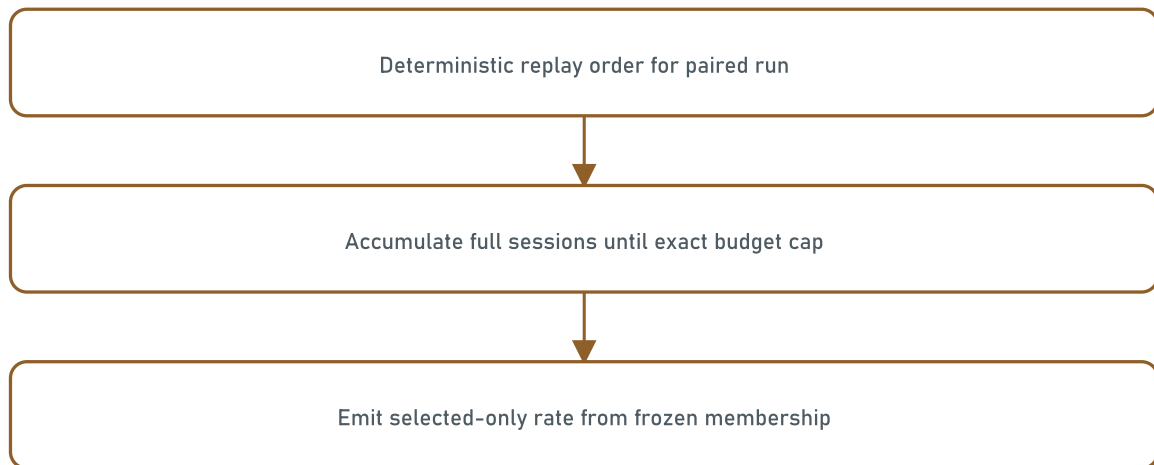
Selection versus estimation: Random, MinHash, and embedding are selection methods under matched exact budgets. Same-agent IDW is a post-freeze estimator applied only to embedding unjudged units.

Overall selection and estimation flow



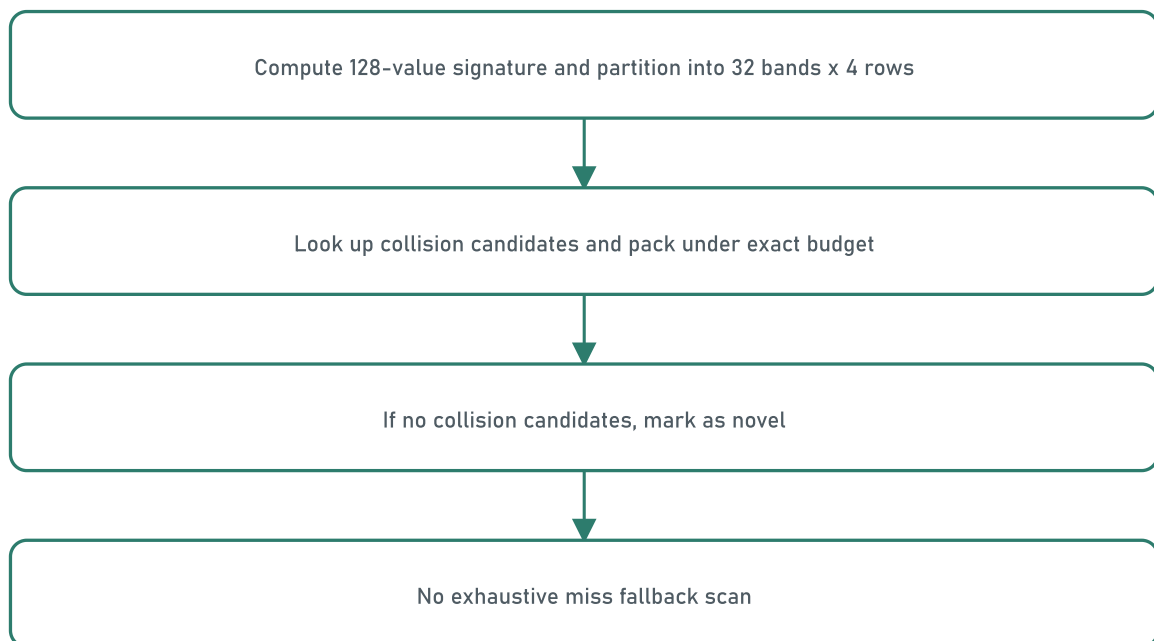
How to interpret it: Top-to-bottom order is execution order. Selection and estimation are intentionally separated to prevent estimator leakage into membership.

Random token-priority explainer



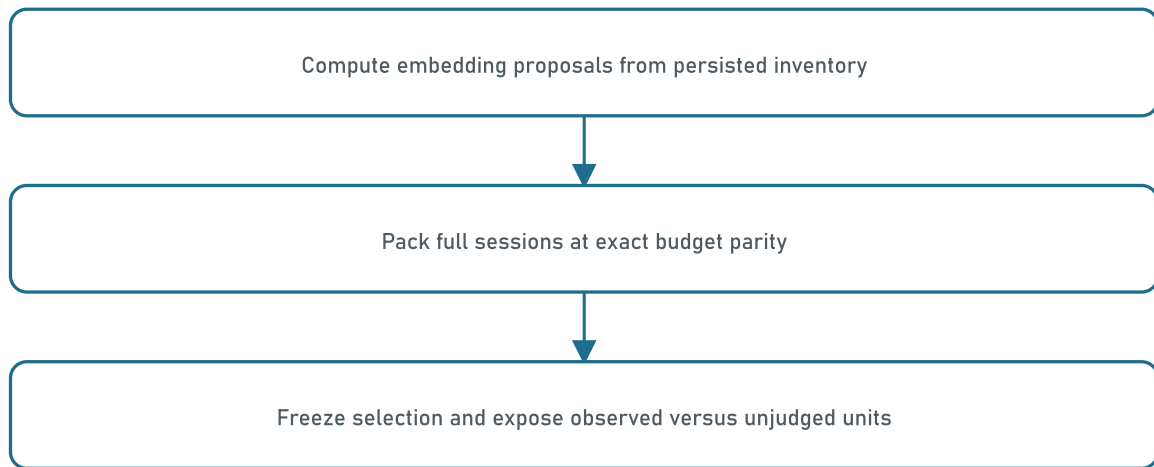
How to interpret it: Random is the simplest accuracy baseline here because it has no adaptive neighbor lookup or post-selection fill policy.

MinHash LSH explainer (128 values, 32x4)



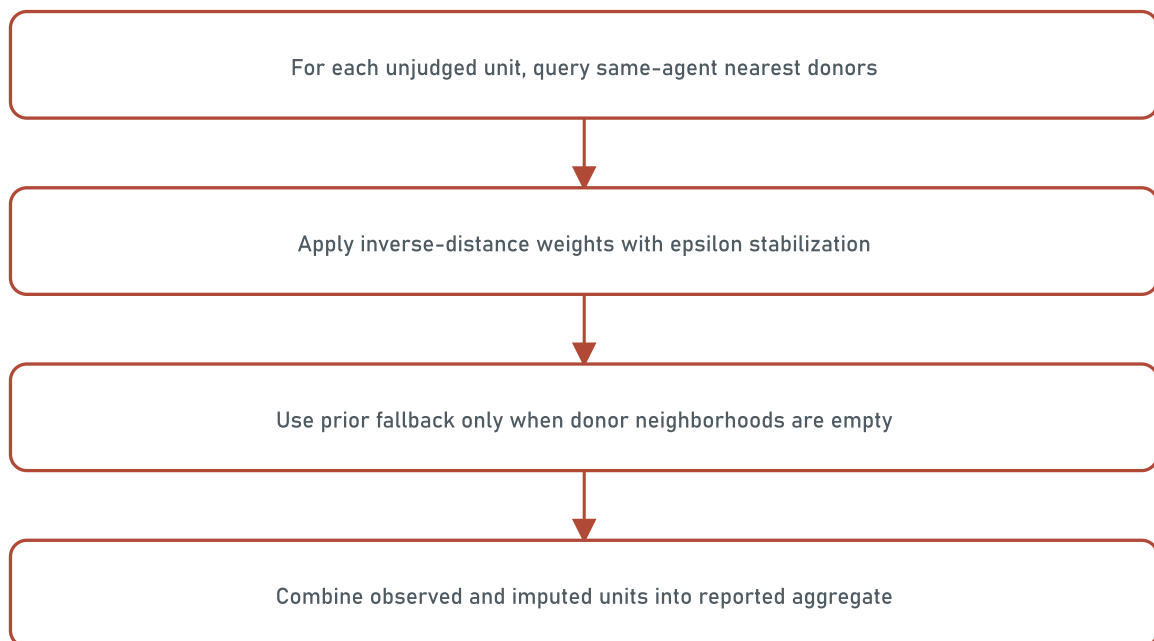
How to interpret it: Bounded collision lookup controls lookup cost. Misses indicate novelty, not a hidden full-corpus rescue path.

Embedding selection explainer



How to interpret it: This diagram covers selection only. Estimation choices are shown separately in IDW.

Post-freeze same-agent IDW explainer



How to interpret it: IDW runs only after selection freeze, so it changes estimation not membership. Treat it as a diagnostic extension, not a selection policy.

What this shows: each method explainer is an inline architecture view of persisted operations; none rely on external assets.

Random token-priority selection

1. Build an exact token budget from persisted budget manifests.
2. Rank sessions deterministically from the paired randomized replay order.
3. Pack maximal whole sessions without splitting.
4. Seal labels until membership is frozen.

Estimation mode	Selected-only
Budget rule	Exact token budget is authoritative
Fill behavior	No separate adaptive fill stage

MinHash LSH 32x4 selection

1. Generate a 128-value MinHash signature split into 32 bands of 4 rows.
2. Propose sessions from candidate collisions under exact budget.
3. If no candidates collide, mark proposal as novel/no-candidate.
4. Do not execute exhaustive full-corpus fallback scans.

LSH shape	128 values; 32 bands x 4 rows
No-candidate handling	Novel proposal; no exhaustive scan
Estimation mode	Selected-only

Embedding selection

1. Embed sessions under the same exact budget process.
2. Select native embedding proposals with deterministic budget fill.
3. Freeze membership before any outcome estimation.
4. Track observed coverage and unjudged units for IDW stage.

Estimation mode	Selected-only baseline + IDW extension
Label handling	Labels remain sealed until membership freeze
Budget packing	Maximal whole-session packing

Same-agent IDW estimation

1. After membership freeze, impute only unjudged units.
2. Use same-agent donor neighborhoods with inverse-distance weighting.
3. Fallback to prior only when no donor is available.
4. Combine observed and imputed means into aggregate rate estimate.

k	8
power	2.000
distance epsilon	0.00000100
prior probability	0.500

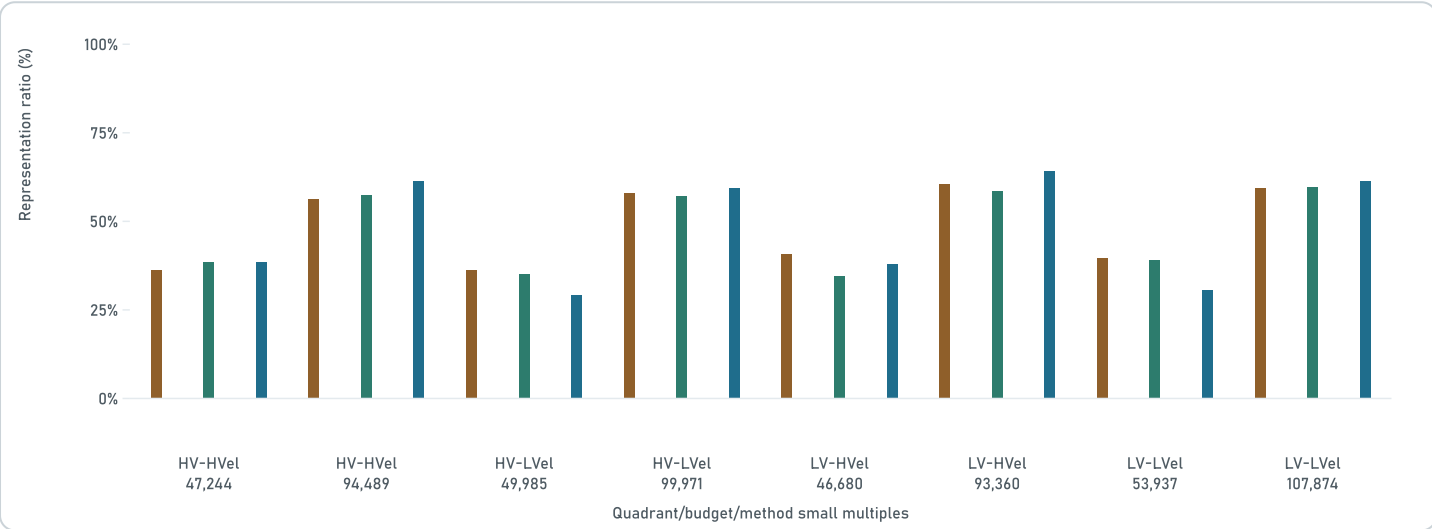
Quadrant Behavior

What this shows: behavior by quadrant under exact-budget parity. Quadrant axes are session variety (low/high) and velocity (low/high). Zero-selection agents are agents with no admitted sessions in a cell.

Quadrant	Sessions	Agents	Source counts
high_variety_high_velocity	713	15	dense_2500:640, historical_300:73
high_variety_low_velocity	687	66	dense_2500:610, historical_300:77
low_variety_high_velocity	687	18	dense_2500:610, historical_300:77
low_variety_low_velocity	713	62	dense_2500:640, historical_300:73

Representation ratio by quadrant, exact budget, and method

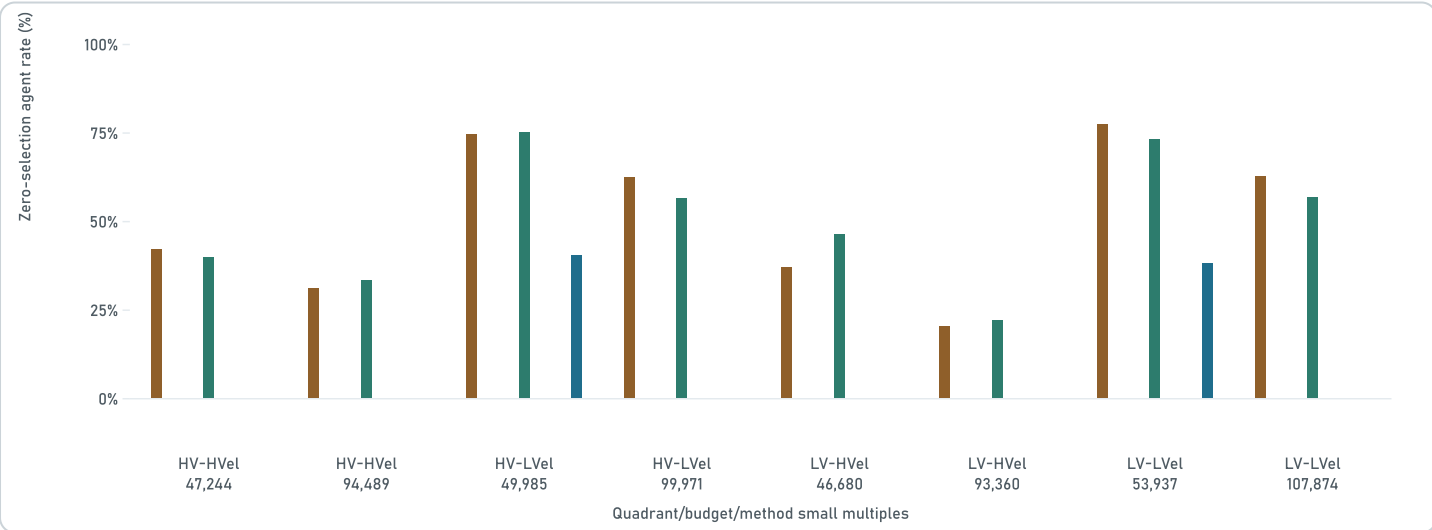
Random selected-only MinHash selected-only Embedding selected-only



How to interpret it: representation range 29.1% to 64.1% across persisted quadrants and methods.

Zero-selection agent rate by quadrant, budget, and method

Random selected-only MinHash selected-only Embedding selected-only



How to interpret it: zero-selection rate range 0.0% to 77.4%; high zero-selection pockets indicate admission sparsity rather than judge latency.

Exact Quadrant Results

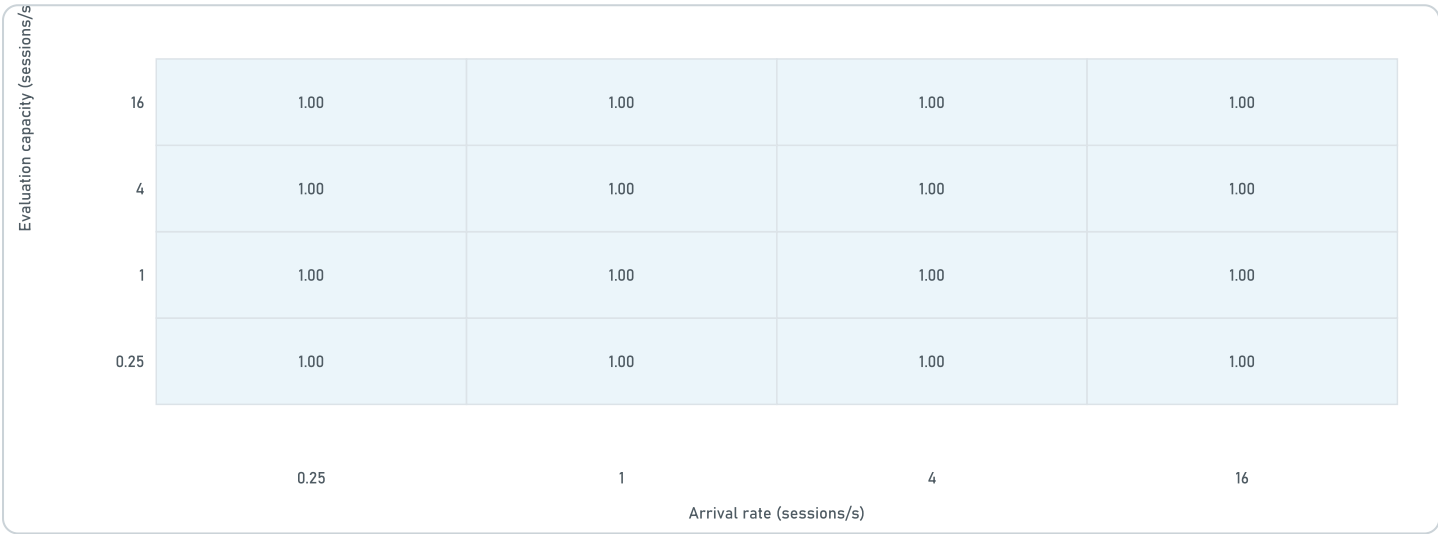
Quadrant	Provenance tier	Exact token budget	Method	Representation	Concept coverage	Zero-selection agents
high_variety_high_velocity	15%	47,244	Embedding selected-only	38.4%	38.4%	0.0%
high_variety_high_velocity	15%	47,244	MinHash selected-only	38.6%	38.6%	40.0%
high_variety_high_velocity	15%	47,244	Random selected-only	36.2%	36.2%	42.2%
high_variety_high_velocity	30%	94,489	Embedding selected-only	61.4%	61.4%	0.0%
high_variety_high_velocity	30%	94,489	MinHash selected-only	57.3%	57.3%	33.3%
high_variety_high_velocity	30%	94,489	Random selected-only	56.3%	56.3%	31.1%
high_variety_low_velocity	15%	49,985	Embedding selected-only	29.1%	29.1%	40.4%
high_variety_low_velocity	15%	49,985	MinHash selected-only	35.1%	35.1%	75.3%
high_variety_low_velocity	15%	49,985	Random selected-only	36.3%	36.3%	74.7%
high_variety_low_velocity	30%	99,971	Embedding selected-only	59.5%	59.5%	0.0%
high_variety_low_velocity	30%	99,971	MinHash selected-only	57.3%	57.3%	56.6%
high_variety_low_velocity	30%	99,971	Random selected-only	58.1%	58.1%	62.6%
low_variety_high_velocity	15%	46,680	Embedding selected-only	37.9%	37.9%	0.0%
low_variety_high_velocity	15%	46,680	MinHash selected-only	34.7%	34.7%	46.3%
low_variety_high_velocity	15%	46,680	Random selected-only	40.6%	40.6%	37.0%
low_variety_high_velocity	30%	93,360	Embedding selected-only	64.1%	64.1%	0.0%
low_variety_high_velocity	30%	93,360	MinHash selected-only	58.5%	58.5%	22.2%
low_variety_high_velocity	30%	93,360	Random selected-only	60.5%	60.5%	20.4%
low_variety_low_velocity	15%	53,937	Embedding selected-only	30.7%	30.7%	38.2%
low_variety_low_velocity	15%	53,937	MinHash selected-only	39.2%	39.2%	73.1%
low_variety_low_velocity	15%	53,937	Random selected-only	39.6%	39.6%	77.4%
low_variety_low_velocity	30%	107,874	Embedding selected-only	61.3%	61.3%	0.0%
low_variety_low_velocity	30%	107,874	MinHash selected-only	59.6%	59.6%	57.0%
low_variety_low_velocity	30%	107,874	Random selected-only	59.4%	59.4%	62.9%

Throughput

What this shows: throughput stress against queue capacity under exact budgets. Token pressure and decision runtime are selection-decision mechanics, not model generation latency.

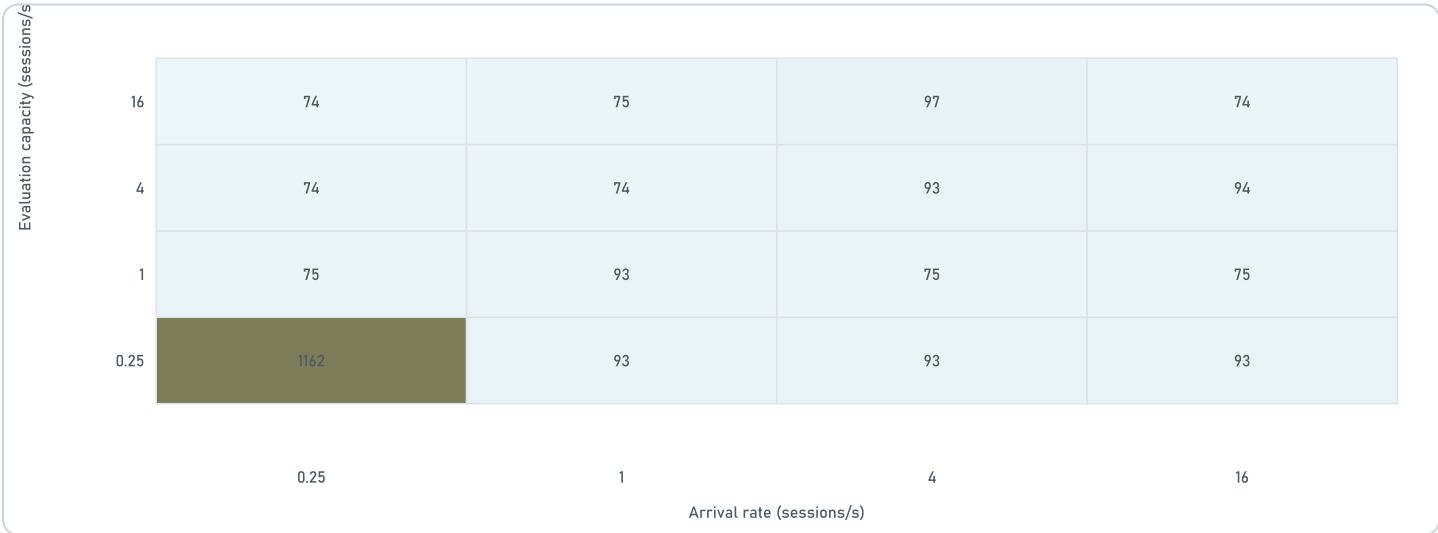
Exact budget	Method	Worst token pressure ratio	Worst p95 decision latency (ms)
97,230	Random selected-only	1.000	31.0
97,230	MinHash selected-only	5.037	4.2
97,230	Embedding selected-only	1.149	533.2
194,460	Random selected-only	1.000	55.1
194,460	MinHash selected-only	2.606	6.0
194,460	Embedding selected-only	1.000	1161.9

Token pressure heatmap (embedding, exact budget 194,460)



How to interpret it: token pressure near 1.0 means sustained budget saturation where arrivals are close to or above effective capacity.

P95 decision latency heatmap (embedding, exact budget 194,460)



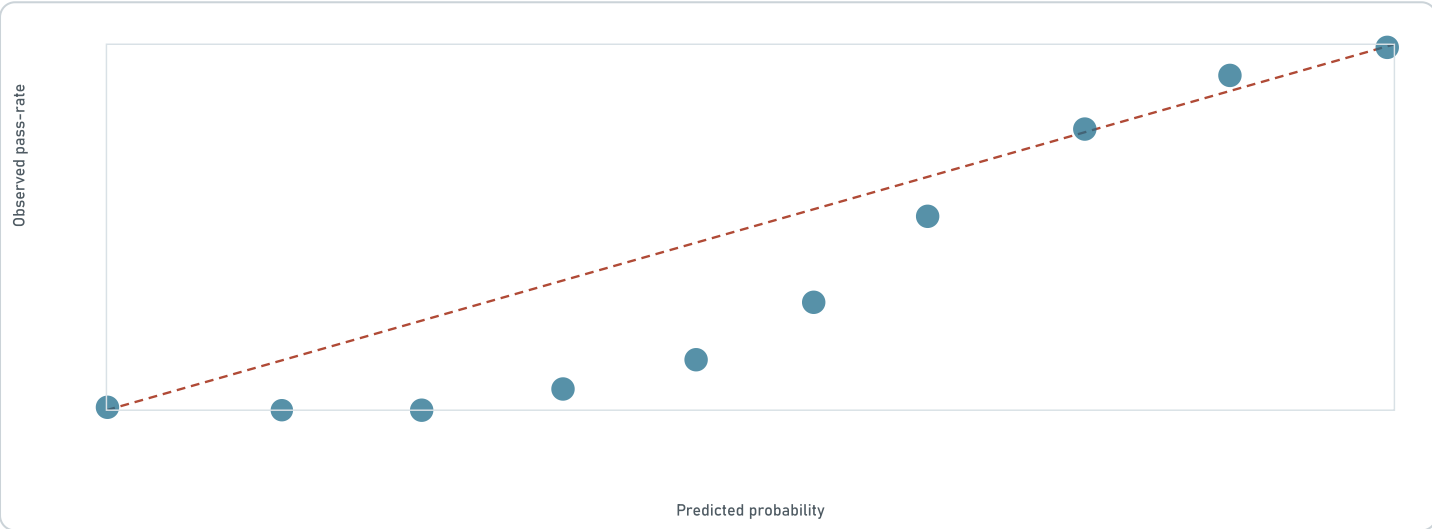
How to interpret it: p95 decision latency peaks align with capacity-constrained cells; this is queueing pressure, not model response time.

Embedding Diagnostics

What this shows: reliability, distance/error behavior, per-agent fit, and provenance composition for a deterministic representative embedding run.

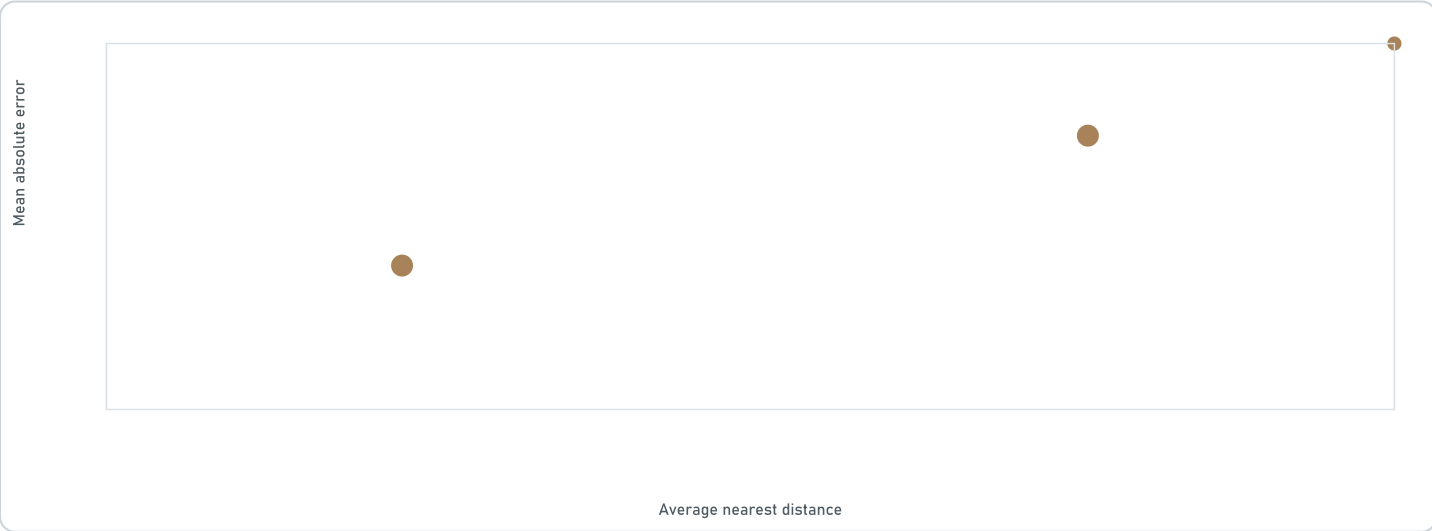
Exact budget	Legacy tier	Repetition	Order hash	Observed	Imputed	Aggregate IDW error	Delta vs selected-only	Unjudged MAE	Unjudged Brier	ECE	LOO MAE	LOO Brier	Zero-donor agents	Prior counts
659,492	50%	0	6017ab8a879a460e119df72ddeee8394a98322f5ca8b04f0ab6665a18b92e87c5	1,279	1,521	0.0301	0.0265	0.2345	0.1120	0.0435	0.2555	0.1267	0	0

Calibration reliability plot (embedding representative run)



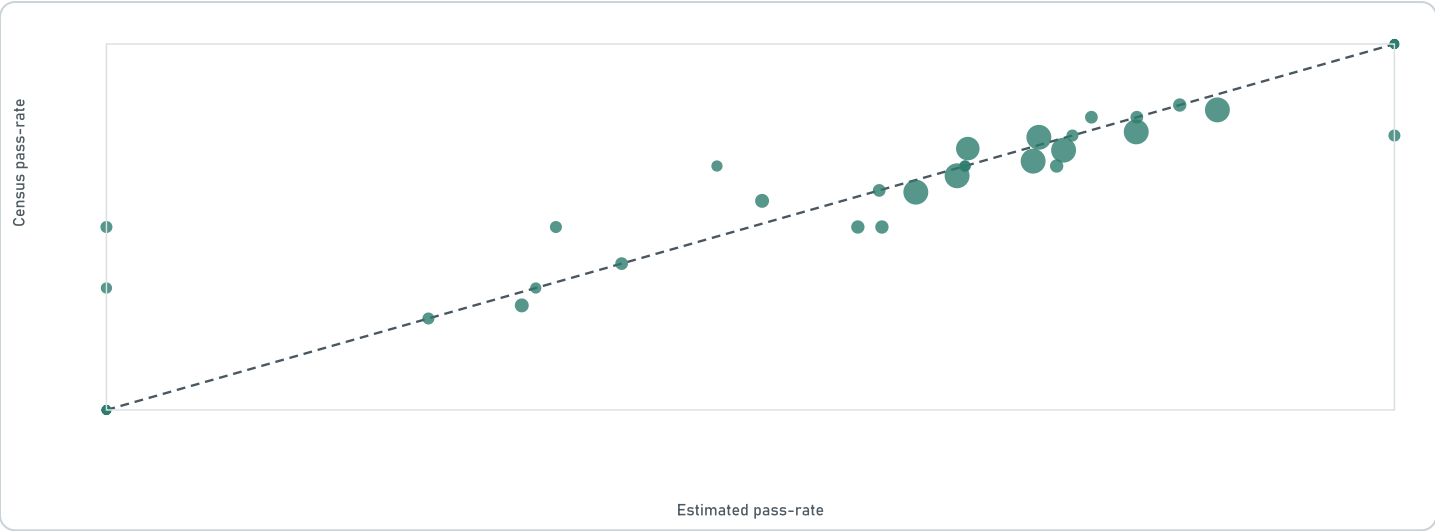
How to interpret it: reliability points near the diagonal are better calibrated; large off-diagonal gaps flag over/under-confidence.

Nearest-distance versus error (embedding representative run)



How to interpret it: rising error with larger nearest distance indicates weaker donor similarity for imputed units.

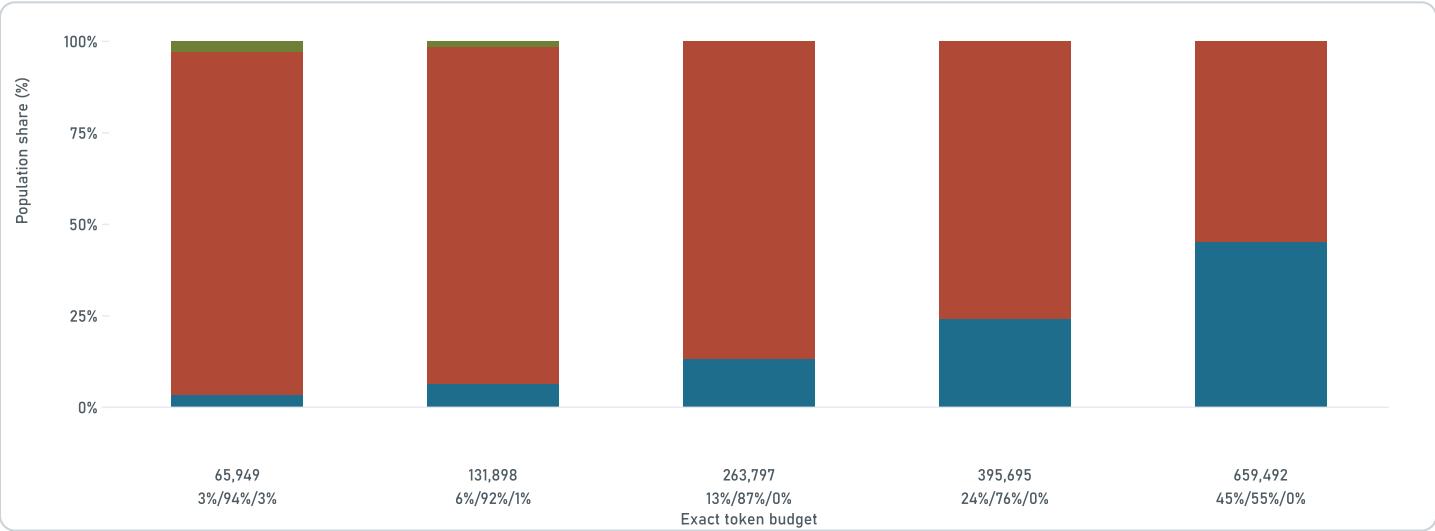
Per-agent estimated versus census dots (embedding representative run)



How to interpret it: per-agent dots near the diagonal indicate accurate agent-level reconstruction; outliers identify agent-specific donor sparsity.

Embedding provenance shares by exact budget (population counts)

■ Observed ■ IDW / exact match ■ Mean / prior fallback



How to interpret it: observed/imputed/fallback shares summarize provenance mix in reported embedding rates; larger fallback share signals lower donor support.

Token and Embedding Ledger

Metric	Value
Token inventory rows	2,800
Emitted tokens	1,318,985
Original tokens	1,318,985
Truncation count	0
Embedding calls	0
Embedding input tokens	0
Embedding model	
Embedding deployment	

Lineage & Integrity

Lineage and integrity records are technical and hash-validated. Reader-facing interpretation is fully contained in this report.

Lineage field	Value
Selection bundle subdir	source_v3
Selection bundle version	sampling-v3-bundle-v1
Selection outcome version	sampling-v3-outcome-v1
Selection manifest version	sampling-v3-manifest-v1
Selection manifest sha256	3ca91469d06f642b4bbb5ada8778ae3a6f19bd9d1dcdabe35904cb758849153f
Selection rerun	False
Augmentation function	augment_v3_outcome_with_idw

Standalone Methodology Summary

- Membership was frozen in the selection-stage bundle before estimation.
- Exact token budgets were preserved cell-for-cell for outcome comparisons.
- Random and MinHash remain selected-only outcome estimators.
- Embedding received post-freeze same-agent IDW estimation over unjudged units.
- IDW parameters: k=8, power=2.0, eps=1e-06, exact_cosine_eps=1e-08, prior=0.5.
- No exhaustive MinHash full-scan fallback was introduced in this report stage.

Reproducibility & Caveats

Reproducibility uses persisted manifests and source hashes with friendly display labels.

Artifact	Relative path	Bytes	SHA-256 prefix
V4 aggregate bundle	aggregate.json	16,149	b588394e48bd5678ee
IDW configuration	idw_config.json	123	b8a70b0295d674ad02
Methodology notes	methodology_delta.md	905	4b7dd760dd9a3afa83
V4 run rows	runs.jsonl	5,135,244	bb25b657eb2c532aef
Selection lineage	source_lineage.json	425	3fea638e4471841565
Selection-stage manifest	source_v3/manifest.json	2,200	3ca91469d06f642b4b

Caveats

- The selection stage did not compute IDW; IDW was applied after membership freeze.
- Adaptive methods are not design-unbiased estimators in this setup.
- Expected labels are pseudo-judge outputs, not human gold labels.
- Synthetic corpus scope may not transfer to production distributions.
- Same-agent donor fallback can increase prior-driven estimates when donor coverage is sparse.
- Only three paired outcome repetitions were run for this bundle.
- Empirical low/high are observed replay ranges, not confidence intervals.